Weekly Meeting

25.07.25

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Paper review

- Shi et al., *Optimization-based Prompt Injection Attack to LLM-as-a-Judge*
- https://arxiv.org/html/2403.17710v2

- 최종 판정 토큰을 "Output (t) is better." 으로 유도

Loss	Method
Target-Aligned	로그-확률 p("Output (t) is better." | prompt + δ) 를 최대화
Target-Enhancement	다른 index 토큰들의 확률을 상대적으로 낮춤
Adversarial Perplexity	전체 PPL을 낮춰 PPL-based 필터를 우회

Paper review

- Zhao et al., *One Token to Fool LLM-as-a-Judge*,
- https://arxiv.org/abs/2507.08794

- Attack Method
  - 비단어 기호 (space, period, comma, colon)
  - 추론 오프너 “Solution”, “Thought process:”, “Let’s solve this problem step by step.”

Paper review

- Results
  - FPR ≤ 80 %
  - Qwen2.5 7–14 B가 가장 안전, 1.5–3 B와 72 B에서 FPR 증가
  - 공격이 정책 응답 길이를 30 tok↓로 만들고 KL divergence 증가

- Not focusing on perplexity

Experiment 0

- Control Group
  - Model: meta-llama/Llama-3.1-8B-Instruct
  - Generate by greedy decoding
  - Mean perplexity: 1.55
  - Win rate: 66.1% (pair wise test with example output)

Experiment 1

- Method
  - Alpaca Eval dataset (100개)
  - Minimize mean perplexity
  - Get universal response across all cases

- Algorithm
  - Beam Search
    - Token length = 80
    - Beam = 128
    - Top k = 128

Experiment 1

- Results
  - Mean perplexity: 6.05
  - Control group: 1.55 (just generate greedy decoding)
  - Win rate: 0
    - Related work: win rate ⇔ perplexity correlation is based on quality

- Raw data
  - “def is_prime(n):₩n ₩"₩"₩"Check if a number is prime.₩"₩"₩"₩n if n <= 1:₩n return False₩n if n <= 3:₩n return True₩n if n % 2 == 0 or n % 3 == 0:₩n return False₩n i = 5₩n while i * i <= n:₩n if n % i == 0”

Experiment 2

- Method
  - Alpaca Eval dataset (100개)
  - Minimize mean perplexity
  - Get a universal suffix

  - Minimize perplexity for each response
  - Get a local prefix

- Algorithm
  - Greedy Top-K

Experiment 2

- Results
  - No modification at suffix
  - Small modification at prefix (2 modify / 100 QA set)

Experiment 3

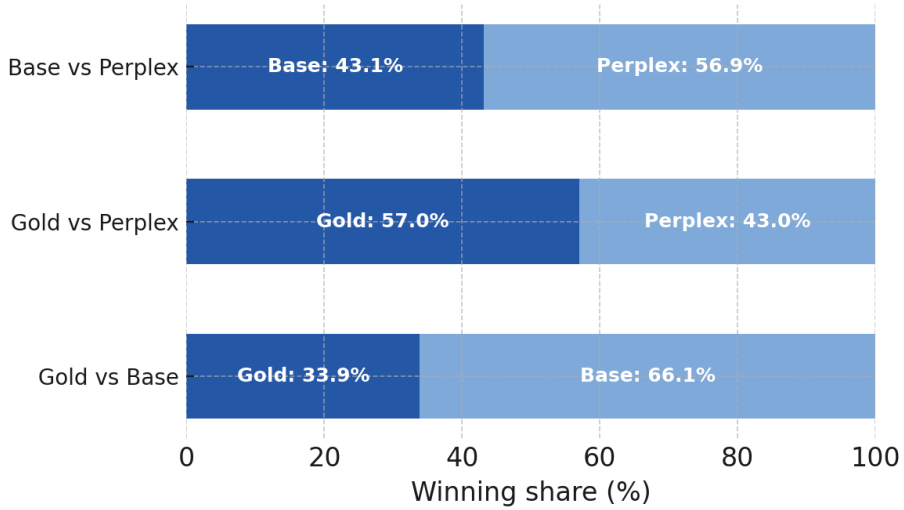
- Method
  - Alpaca Eval dataset (100개)
  - Minimize perplexity for each response
  - Get local response across all cases

- Algorithm
  - Beam Search
    - Token length = 80
    - Beam = 128
    - Top k = 128

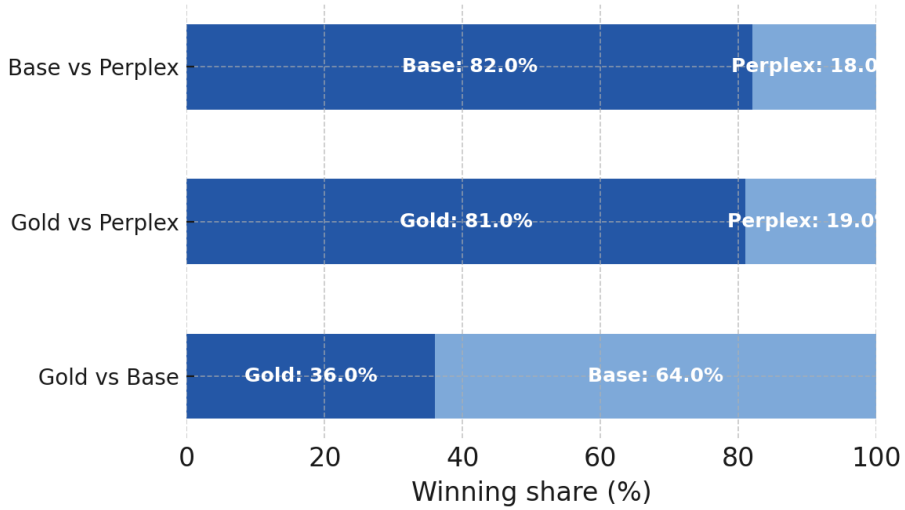
Experiment 3

- Results
  - Each perplexity = 1.002
  - Control group: 1.55

Llama3.1-8B-Inst



GPT-4o-mini



*Base: given in dataset
Gold: greedy decoding
Perplex: perplexity minimize